

Banegas’ Second Family of Irreducible Pentanomials: a Closed-Form Reduction with $3m - c - 3$ XOR Gates

Marcos Tomaszewski
Ricardo Felipe Custódio

Laboratório de Segurança em Computação (LabSEC)
Departamento de Informática e Estatística, Universidade Federal de Santa Catarina
Florianópolis, SC, Brazil

Abstract

In a polynomial basis, the cost of multiplying in $\mathbb{F}_{2^m}$ is the cost of a carry-less product plus the cost of reducing it modulo the defining polynomial, and only the second part depends on the choice of the polynomial. Searching all irreducible pentanomials of small degree with a greedy XOR counter, Banegas found two families, $x^{2b+c} + x^{b+c} + x^b + x^c + 1$ and $x^{b+2c} + x^{b+c} + x^b + x^c + 1$. The first was analysed by Banegas, Custódio and Panario, who proved a reduction cost of $3m - 2$ XORs; the second was left as future work. This paper completes that programme. For $f = x^{b+2c} + x^{b+c} + x^b + x^c + 1$, $m = b + 2c$, $b > c \geq 1$, we derive a closed form for the quotient of the division by f in terms of “comb” sums of stride $3c$, and turn it into an explicit reduction program with at most $3m - c - 3$ XORs and depth at most $\lfloor (m - 2)/3c \rfloor + 4$, valid for every b and c . The resulting multiplier uses $m^2 + m - c - 2$ XORs. Since c may be as large as $(m - 1)/3$, the cost approaches $\frac{8}{3}m$. We verified the program symbolically for all 14 850 members with $m \leq 300$. It is never worse than the greedy count on the 3 541 generic members with $m \leq 150$, is strictly better on 1 271 of them, and for $b < 2c$ it improves the greedy count by exactly $2c - b$. Over the degrees $5 \leq m \leq 1100$, the family has irreducible members in 720 degrees, including 288 degrees without irreducible trinomials; wherever both families exist, the second one is cheaper in 403 of 405 degrees, by 7.3% on average. At the NIST degrees 163 and 571 its best members reduce with 440 and 1528 XORs, against 487 and 1711 for the first family and 571 and 2003 reported for the NIST pentanomials.

Keywords: binary fields, irreducible pentanomials, modular reduction, XOR count, bit-parallel multipliers, polynomial basis.

1 Introduction

Finite field arithmetic. Finite fields are the algebraic substrate of most cryptographic primitives and of many error-correcting codes [15, 19]. Two kinds of fields dominate implementations. *Prime fields* $\mathbb{F}_p$ with a large prime p use multiprecision integer arithmetic, and special primes make modular reduction cheap. *Binary extension fields* $\mathbb{F}_{2^m} \cong \mathbb{F}_2[x]/(f)$, with f irreducible of degree m , use carry-free arithmetic: addition is a bitwise XOR and multiplication is a carry-less product followed by a reduction modulo f . Elements can be represented in polynomial, normal, shifted or Montgomery bases, or as towers of subfields. In the polynomial basis the product stage does not depend on f , while the reduction depends only on f ; choosing f is therefore a design decision with a direct effect on area, delay and energy in hardware and on the instruction count of bitsliced software.

Prime fields in standards, binary fields in practice. For elliptic curve cryptography the NIST recommendations moved to prime fields: SP 800-186 keeps the prime-field curves, adds Edwards curves, and deprecates the curves over binary fields, whose defining polynomials were the trinomials of degrees 233 and 409 and the pentanomials of degrees 163, 283 and 571 [9, 22].

Binary fields in polynomial basis nevertheless remain in daily use: the AES operates in $\mathbb{F}_{2^8}$ [21]; GCM authenticates with a pentanomial of degree 128 [12, 14]; Classic McEliece computes in $\mathbb{F}_{2^{12}}$ and $\mathbb{F}_{2^{13}}$ [1, 6]; the post-quantum signature scheme FAEST works in $\mathbb{F}_{2^\lambda}$ [5]; succinct arguments use towers of binary fields [11]; and Reed–Solomon and BCH codecs, the motivation of the classic work of Paar [23], still reduce in binary fields. In all these settings each XOR gate counts.

Special polynomials. Trinomials give the cheapest reduction, $2m - 2$ XORs [29, 32], but no irreducible trinomial exists for many degrees: none for $m \equiv 0 \pmod{8}$ and none for many odd $m \equiv \pm 3 \pmod{8}$, as a consequence of Swan’s theorem [30]. For those degrees one uses pentanomials, and a line of work proposed pentanomial families with low reduction or multiplication cost [2, 10, 16, 17, 25, 26].

Banegas’ two families. In his master’s thesis at LabSEC/UFSC, Banegas implemented a greedy XOR counter, *Conta-XOR*, ran it over all irreducible pentanomials of degrees 19 and 163, and identified two families of pentanomials with small counts [3]:

$$\text{(first)} \quad x^{2b+c} + x^{b+c} + x^b + x^c + 1, \quad \text{(second)} \quad x^{b+2c} + x^{b+c} + x^b + x^c + 1.$$

The first family was analysed by Banegas, Custódio and Panario (BCP): the reduction costs $3m - 2$ XORs, and $\frac{12}{5}m - 1$ when $b = 2c$ [2]. About the second, the thesis states that it “demonstrated that a greater understanding of the reduction process that involves it is necessary” (our translation) and proposes its study as future work [3, Sec. 4.5]. Ten years later, this paper carries out that study.

Contributions. Results marked $\blacklozenge$ are new; every other result is cited.

1. $\blacklozenge$ Structural facts about the second family, including the identity $(x^c + 1)f = x^{b+3c} + x^b + x^{2c} + 1$, which drives everything else (Section 3).
2. $\blacklozenge$ A closed form for the quotient $\lfloor D/f \rfloor$ in terms of comb sums of stride $3c$ (Theorem 5).
3. $\blacklozenge$ An explicit reduction program with at most $3m - c - 3$ XORs and depth at most $\lfloor (m-2)/3c \rfloor + 4$, for all $b > c \geq 1$, and the corresponding multiplier with $m^2 + m - c - 2$ XORs (Theorem 6, Proposition 7, and Corollary 8).
4. $\blacklozenge$ A word-level formulation with $O(\log(m/c))$ shift-and-XOR operations, and a low-depth gate-level variant (Section 6).
5. $\blacklozenge$ A comparison with the Conta-XOR count: the closed form is never worse on generic members, it explains the counts reported in the thesis, and it improves on them by exactly $2c - b$ when $b < 2c$ (Section 7).
6. $\blacklozenge$ A census of irreducible members up to degree 1100 and a comparison with the first family, with trinomials and with the NIST pentanomials (Section 8).

2 Background

Everything in this section is from the literature.

2.1 Reduction and its cost

Let $f = x^m + g$ with $g(0) = 1$ and $\deg g < m$. The product of two elements of $\mathbb{F}_{2^m}$ is $D = \sum_{j=0}^{2m-2} d_j x^j$, which we split as

$$D = L + x^m H, \quad L = \sum_{i=0}^{m-1} d_i x^i, \quad H = \sum_{s=0}^{m-2} h_s x^s, \quad h_s = d_{m+s}. \quad (1)$$

Table 1: Reduction costs of special irreducible polynomials (literature).

Polynomial	Condition	XORs for reduction	Ref.
trinomial $x^m + x^k + 1$	$k \leq m/2$	$2m - 2$	[29, 32]
equally spaced $x^{4c} + x^{3c} + x^{2c} + x^c + 1$	$m = 4c$	$\frac{7}{4}m - 1$	[3, 16]
Banegas' first family $x^{2b+c} + x^{b+c} + x^b + x^c + 1$	$b > c$	$3m - 2$	[2]
the same, with $b = 2c$	$m = 5c$	$\frac{12}{5}m - 1$	[2]

Reduction computes $R = D \bmod f$. It is a fixed $\mathbb{F}_2$ -linear map: the *reduction matrix* $R_f \in \mathbb{F}_2^{m \times (2m-1)}$ has as column j the coefficients of $x^j \bmod f$, and $R = R_f(d_0, \dots, d_{2m-2})^\top$. A *linear straight-line program* computes R with instructions $t \leftarrow u \oplus v$ over inputs and earlier results; its length is the number of XOR gates, and its *depth* is the length of the longest chain of dependent gates. The *direct count* $\text{dXOR}(f) = \sum_i (\text{weight of row } i - 1)$ adds each output from scratch. We write $\text{Red}(f)$ for the minimum length of a program for R_f . Computing $\text{Red}(f)$ is NP-hard in general [7]; heuristics include the greedy algorithm of Paar, which repeatedly replaces the most frequent pair of operands by a new variable [23], the heuristic of Boyar and Peralta, which allows cancellations [8], and their many refinements [4, 18, 28, 31, 33]. Exact values are obtained with SAT solvers for small instances [13, 27].

2.2 Conta-XOR

Conta-XOR [3, Ch. 4] builds a reduction matrix by repeated substitution of x^m by g , removes pairs of equal terms in each output, replaces repeatedly the most repeated pair of operands by a temporary variable, and reports the number of temporaries plus the remaining additions. We write $\text{CX}(f)$ for its value, which we recomputed with an independent implementation that reproduces the values published in [3]; ties between equally frequent pairs are broken lexicographically.

2.3 Known special pentanomials

Table 1 lists closed-form reduction costs from the literature for the special polynomials we compare with.

For a schoolbook product the partial products cost m^2 AND gates and $(m-1)^2$ XOR gates, so a multiplier costs $(m-1)^2$ plus the reduction; for the first family this is $m^2 + m - 1$ XORs [2].

3 The second family

Definition 1. For integers $b > c \geq 1$ let $m = b + 2c$ and

$$f_{b,c}(x) = x^{b+2c} + x^{b+c} + x^b + x^c + 1.$$

We call $\{f_{b,c}\}$ *Banegas' second family*, and write $n = m - 2$ for the degree of H in (1).

The constraint $b > c$ is part of the definition in [3]; it also guarantees that the exponents are distinct, because $b + c < b + 2c$ and $c < b$.

Proposition 2 (♦). *Let $f = f_{b,c}$.*

- (i) $x^m \equiv (x^b + 1)(x^c + 1) \pmod{f}$.
- (ii) $(x^c + 1)f = x^{b+3c} + x^b + x^{2c} + 1$, hence $x^b(x^{3c} + 1) \equiv (x^c + 1)^2 \pmod{f}$.
- (iii) $1 \leq c \leq \lfloor (m-1)/3 \rfloor$ and $b \equiv m \pmod{2}$.
- (iv) If $g = \gcd(b, c) > 1$ then $f(x) = G(x^g)$ for the pentanomial $G = f_{b/g, c/g}$.
- (v) For $b = 2c$, f is the equally spaced pentanomial $x^{4c} + x^{3c} + x^{2c} + x^c + 1$.

Proof. (i) $x^{b+c} + x^b + x^c + 1 = (x^b + 1)(x^c + 1)$. (ii) Expanding, $(x^c + 1)(x^{b+2c} + x^{b+c} + x^b + x^c + 1) = x^{b+3c} + x^{b+2c} + x^{b+2c} + x^{b+c} + x^{b+c} + x^b + x^{2c} + x^c + x^c + 1$, and the repeated terms cancel. Reducing $x^{b+3c} = x^{2c} \cdot x^m$ gives the second statement. (iii) $b > c$ and $m = b + 2c$ give $m > 3c$; $m - b = 2c$ is even. (iv) and (v) are immediate. $\square$

Identity (ii) is the key to the whole analysis: a shift by $3c$ applied to the high part can be undone at the price of two low-order terms. The first family has the analogous identity $(x^b + 1)f = x^{3b+c} + x^{2b} + x^c + 1$, obtained in the same way, but with a shift of b instead of c ; this is one reason why the two families behave differently.

4 A closed form for the quotient

Lemma 3 (Division identities). *Let $f = x^m + \sum_{e \in S} x^e + 1$, $S \subseteq \{1, \dots, m-1\}$, and let $Q = \lfloor D/f \rfloor = \sum_{j=0}^n q_j x^j$. Then*

$$h_j = q_j + \sum_{e \in S} q_{j+m-e}, \quad 0 \leq j \leq n, \quad (2)$$

$$r_i = d_i + \sum_{e \in S \cup \{0\}} q_{i-e}, \quad 0 \leq i \leq m-1, \quad (3)$$

with $q_k = 0$ for $k < 0$ or $k > n$.

Proof. This is polynomial long division written coefficientwise: from $D = Qf + R$ and $f = x^m + g$, compare the coefficients of x^{m+j} ($j \geq 0$), where R does not contribute, and of x^i ($i < m$), where $x^m Q$ does not contribute. $\square$

For the second family $S = \{c, b, b+c\}$ and $m - S = \{b+c, 2c, c\}$, so (2) reads

$$h_j = q_j + q_{j+c} + q_{j+2c} + q_{j+b+c}. \quad (4)$$

Equation (4) determines Q from the top down, but evaluating it directly costs almost one XOR per coefficient of Q . The next theorem solves the recurrence.

Definition 4 ($\blacklozenge$ Comb sums). For $t \geq 0$ let

$$C_t = \sum_{k \geq 0, t+3ck \leq n} h_{t+3ck},$$

the sum of the high coefficients whose index is congruent to t modulo $3c$ and at least t ; $C_t = 0$ for $t > n$. Equivalently $C_t = h_t \oplus C_{t+3c}$.

Theorem 5 ($\blacklozenge$ Closed-form quotient). *For $f = f_{b,c}$ and $0 \leq j \leq n$,*

$$q_j = C_j + C_{j+c} + h_{j+b+c}, \quad (5)$$

where $h_s = 0$ for $s > n$. Consequently the polynomial $U = (1+x^c)Q$, with coefficients $U_k = q_k + q_{k-c}$, satisfies

$$U_k = \begin{cases} C_k + C_{k+c} + h_{k+b+c}, & 0 \leq k < c, \\ C_{k-c} + C_{k+c} + h_{k+b}, & k \geq c, \end{cases} \quad (6)$$

and the remainder is

$$r_i = d_i + U_i + U_{i-b} \quad (0 \leq i \leq m-1; U_{i-b} = 0 \text{ if } i < b). \quad (7)$$

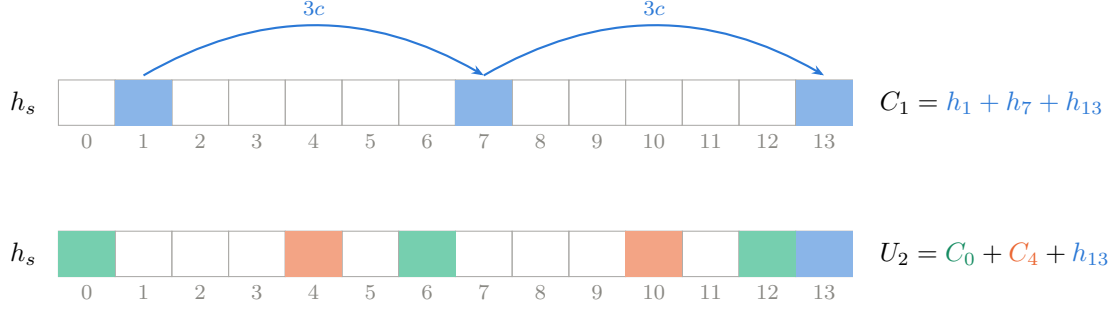


Figure 1: ♦ Comb sums and the closed form (6), for $b = 11$, $c = 2$ ($m = 15$, $n = 13$, stride $3c = 6$). Top: the comb sum C_1 . Bottom: $U_2 = C_0 + C_4 + h_{13}$, the sum of two comb sums and one extra high coefficient ($k = 2 \geq c$, so $C_{k-c} = C_0$, $C_{k+c} = C_4$ and $h_{k+b} = h_{13}$).

Proof. Define $\tilde{q}_j$ by the right-hand side of (5) for all $j \geq 0$. Then $\tilde{q}_j = 0$ for $j > n$, since every index involved exceeds n . We show that $\tilde{q}$ satisfies (4); since (4) determines q_j uniquely from $q_{j+1}, \dots, q_n$, it follows that $\tilde{q} = q$. Adding the four right-hand sides,

$$\begin{aligned} \tilde{q}_j + \tilde{q}_{j+c} + \tilde{q}_{j+2c} + \tilde{q}_{j+b+c} &= (C_j + C_{j+c} + h_{j+b+c}) + (C_{j+c} + C_{j+2c} + h_{j+b+2c}) \\ &\quad + (C_{j+2c} + C_{j+3c} + h_{j+b+3c}) + (C_{j+b+c} + C_{j+b+2c} + h_{j+2b+2c}). \end{aligned}$$

Since $b + 2c = m > n$, the terms h_{j+b+2c} , h_{j+b+3c} , $h_{j+2b+2c}$ and C_{j+b+2c} vanish. The pairs C_{j+c} and C_{j+2c} cancel. Finally $C_{j+b+c} = h_{j+b+c}$, because $j + b + c + 3c > n$; both sides are zero when $j + b + c > n$. The sum therefore equals $C_j + C_{j+3c} = h_j$, which is (4).

For (6): if $k < c$ then $q_{k-c} = 0$ and $U_k = q_k$. If $k \geq c$, then $U_k = q_k + q_{k-c} = (C_k + C_{k+c} + h_{k+b+c}) + (C_{k-c} + C_k + h_{k+b})$; the term h_{k+b+c} vanishes because $k + b + c \geq b + 2c > n$, and C_k cancels. Finally (3) with $S = \{c, b, b + c\}$ gives $r_i = d_i + (q_i + q_{i-c}) + (q_{i-b} + q_{i-b-c}) = d_i + U_i + U_{i-b}$. □

5 The closed-form reduction program

Theorem 5 suggests computing the comb sums once, sharing them among all U_k , and sharing each U_k between the two outputs r_k and r_{k+b} that use it. One more saving is available: for $i \leq c - 2$, the pair $C_i + h_{i+b+c}$ appears both in U_i and in U_{i+c} , because $h_{(i+c)+b} = h_{i+b+c}$.

Theorem 6 (♦ Main result). *For all $b > c \geq 1$, Algorithm 1 computes $D \bmod f_{b,c}$ with at most $3m - c - 3$ XOR gates. Hence $\text{Red}(f_{b,c}) \leq 3m - c - 3$.*

Proof. Correctness. The values C_t are the comb sums. For $k \leq c - 2$, $Z_k \oplus C_{k+c} = C_k + C_{k+c} + h_{k+b+c} = U_k$ by (6). For $k = c - 1$ the extra term $h_{c-1+b+c} = h_{m-1}$ vanishes. For $c \leq k \leq 2c - 2$, $Z_{k-c} \oplus C_{k+c} = C_{k-c} + h_{k+b} + C_{k+c} = U_k$. For $2c - 1 \leq k \leq b + c - 2$ we have $k + b > n$, so $U_k = C_{k-c} + C_{k+c}$; for $k \geq b + c - 1$ also $k + c > n$ and $U_k = C_{k-c}$. The outputs are (7).

Count. The comb sums cost one XOR for each t with $t + 3c \leq n$, that is $\max(0, n - 3c + 1) = \max(0, b - c - 1)$. The pairs Z_i cost $c - 1$. The U_k cost one XOR each for $0 \leq k \leq b + c - 2$ and nothing afterwards: $b + c - 1$. The outputs cost m XORs for $d_i \oplus U_i$ and $m - b = 2c$ more for the U_{i-b} . Since $b > c$,

$$(b - c - 1) + (c - 1) + (b + c - 1) + m + 2c = 2b + 3c - 3 + m = 3m - c - 3.$$

If some operands coincide or cancel, which can happen only for special ratios b/c , the program becomes shorter, never longer. □

Proposition 7 (♦ Depth). *The XOR depth of Algorithm 1 is at most $\lfloor n/3c \rfloor + 4$. In particular it is at most 5 when $b \leq 4c + 1$.*

Algorithm 1: ♦ Reduction modulo $f_{b,c}$, $m = b + 2c$, $n = m - 2$, $h_s = d_{m+s}$.

Input : $d_0, \dots, d_{2m-2}$
Output : $r_0, \dots, r_{m-1}$ with $\sum r_i x^i = D \bmod f_{b,c}$

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1 for  $t \leftarrow n$  to 0 do                                     // comb sums,  $b - c - 1$  XORs
2   if  $t + 3c \leq n$  then  $C_t \leftarrow h_t \oplus C_{t+3c}$ ;
3   else  $C_t \leftarrow h_t$ ;
4 for  $i \leftarrow 0$  to  $c - 2$  do                               // shared pairs,  $c - 1$  XORs
5    $Z_i \leftarrow C_i \oplus h_{i+b+c}$ 
6 for  $k \leftarrow 0$  to  $m - 1$  do                             //  $b + c - 1$  XORs in total
7   if  $k \leq c - 2$  then
8      $U_k \leftarrow Z_k \oplus C_{k+c}$ 
9   else if  $k = c - 1$  then
10     $U_k \leftarrow C_{c-1} \oplus C_{2c-1}$ 
11  else if  $k \leq 2c - 2$  then
12     $U_k \leftarrow Z_{k-c} \oplus C_{k+c}$ 
13  else if  $k \leq b + c - 2$  then
14     $U_k \leftarrow C_{k-c} \oplus C_{k+c}$ 
15  else  $U_k \leftarrow C_{k-c}$ ;
16 for  $i \leftarrow 0$  to  $m - 1$  do                             //  $m + (m - b)$  XORs
17   if  $i \geq b$  then  $r_i \leftarrow (d_i \oplus U_i) \oplus U_{i-b}$ ;
18   else  $r_i \leftarrow d_i \oplus U_i$ ;
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Proof. C_t is a chain of $\lfloor (n - t)/3c \rfloor$ gates, so its depth is at most $\delta = \lfloor n/3c \rfloor$. Then Z_i has depth at most $\delta + 1$, U_k at most $\delta + 2$, and r_i at most $\delta + 4$. If $b \leq 4c + 1$ then $n = b + 2c - 2 < 6c$ and $\delta \leq 1$. $\square$

Over the 14748 pairs with $7 \leq m < 300$ the measured depth is $\delta + 3$ in 97% of the cases. When b is much larger than c the chain of comb sums grows; computing the comb sums with a balanced tree reduces the depth to $O(\log(n/c))$ at the price of more gates.

Corollary 8 (♦ Multiplier). *A schoolbook multiplier for $\mathbb{F}_2[x]/(f_{b,c})$ needs m^2 AND gates and at most $(m - 1)^2 + 3m - c - 3 = m^2 + m - c - 2$ XOR gates, that is, $c + 1$ fewer XORs than the first family (Table 1).*

Example 9 (♦). Degree $16 \equiv 0 \pmod{8}$ has no irreducible trinomial. The pentanomial $f_{6,5} = x^{16} + x^{11} + x^6 + x^5 + 1$ is irreducible. Here $n = 14$ and $3c = 15 > n$, so every comb sum is a single coefficient, $C_t = h_t$, and Algorithm 1 reads:

step	instructions	XORs
pairs	$Z_i = h_i \oplus h_{i+11}$, $i = 0, \dots, 3$	4
U	$U_i = Z_i \oplus h_{i+5}$ ($i \leq 3$); $U_4 = h_4 \oplus h_9$	5
	$U_i = Z_{i-5} \oplus h_{i+5}$ ($5 \leq i \leq 8$); $U_9 = h_4 \oplus h_{14}$	5
	$U_i = h_{i-5}$ for $10 \leq i \leq 15$	0
	outputs $r_i = d_i \oplus U_i$ ($0 \leq i \leq 5$); $r_i = d_i \oplus U_i \oplus U_{i-6}$ ($6 \leq i \leq 15$)	26
total		$40 = 3m - c - 3$

For comparison, $\text{dXOR}(f) = 54$ and $\text{CX}(f) = 44$.

6 Word-level formulation and a low-depth variant

Theorem 5 has a compact form in terms of whole polynomials. Write $A \gg k$ for $\lfloor A/x^k \rfloor$ and $A \ll k$ for $x^k A$.

Corollary 10 ($\blacklozenge$ Word-level reduction). *With L and H as in (1),*

$$\begin{aligned} C &= \sum_{k \geq 0} (H \gg 3ck), & Q &= C \oplus (C \gg c) \oplus (H \gg (b+c)), \\ U &= Q \oplus (Q \ll c), & D \bmod f_{b,c} &= (L \oplus U \oplus (U \ll b)) \bmod x^m. \end{aligned}$$

The sum defining C can be evaluated by doubling: $C \leftarrow H$, then $C \leftarrow C \oplus (C \gg 3c \cdot 2^j)$ for $j = 0, 1, \dots$ while $3c \cdot 2^j \leq n$.

Proof. The coefficient of x^t in $\sum_k (H \gg 3ck)$ is C_t ; the formula for Q is (5); $U = (1 + x^c)Q$; and $r_i = d_i + U_i + U_{i-b}$ is (7). After the doubling step j , the coefficient t of C is the sum of h_{t+3ck} for $0 \leq k < 2^{j+1}$, which gives the comb sums once $3c \cdot 2^{j+1} > n$. $\square$

In software the reduction therefore costs $s + 5$ shift-and-XOR operations on m -bit vectors, where $s = \lceil \log_2((n+1)/3c) \rceil$ doubling steps (and $s = 0$ if $3c > n$), independently of the word size. For the members with $b \leq 4c + 1$, which include the best members at $m = 163$ and 571 , $s \leq 1$ and the whole reduction takes at most six shift-and-XOR operations.

The same doubling gives a gate-level program of logarithmic depth.

Proposition 11 ($\blacklozenge$ Low-depth variant). *Replacing the chain of comb sums in Algorithm 1 by the doubling of Corollary 10 gives a program with at most*

$$3m - c - 3 + \sum_{j=1}^{s-1} \max(0, n + 1 - 3c \cdot 2^j)$$

XOR gates and depth at most $s + 4$.

Proof. Doubling step j costs one gate for each t with $t + 3c \cdot 2^j \leq n$, that is $\max(0, n + 1 - 3c \cdot 2^j)$ gates, and adds one level of depth. Step $j = 0$ costs $n + 1 - 3c = b - c - 1$ gates, exactly the cost of the chain in Algorithm 1; the other steps are the extra term. The rest of the program is unchanged and adds at most 4 levels, as in the proof of Proposition 7. $\square$

We verified the variant symbolically for all members with $m < 260$. When $b \leq 4c + 1$ both programs coincide. For the member $x^{571} + x^{549} + x^{527} + x^{22} + 1$ ($c = 22$), the chain has depth 11 with 1688 gates, and the doubling has depth 7 with 2474 gates, which illustrates the area-delay trade-off when c is small.

7 Comparison with the greedy count

The second family was discovered through Conta-XOR, so the first question is how the closed form relates to the greedy counts. We call a member *special* if $b/c \in \{\frac{3}{2}, 2, 3, 4, 5\}$; then $f(x) = G(x^g)$ with $g \geq c/2$ and G a pentanomial of degree at most 7 (Proposition 2(iv)), and $b = 2c$ is the equally spaced case. All other members are *generic*.

Observation 12 ($\blacklozenge$). *We computed $CX(f_{b,c})$, and the best of 20 additional greedy runs with random tie-breaking in the spirit of [31], for all 3674 pairs with $5 \leq m \leq 150$. Among the 3541 generic members:*

- (a) *the closed form is never longer than either greedy value;*
- (b) *for all 879 members with $b < 2c$, $CX(f) = 2m + 3c - 3$, so the closed form is shorter by exactly $2c - b$;*
- (c) *for all 1312 members with $2c < b < 6c$, $CX(f) = 3m - c - 3$, equal to the closed form;*
- (d) *for $b \geq 6c$ the greedy count exceeds the closed form in a growing fraction of cases (392 of 1350), by up to 112 XORs.*

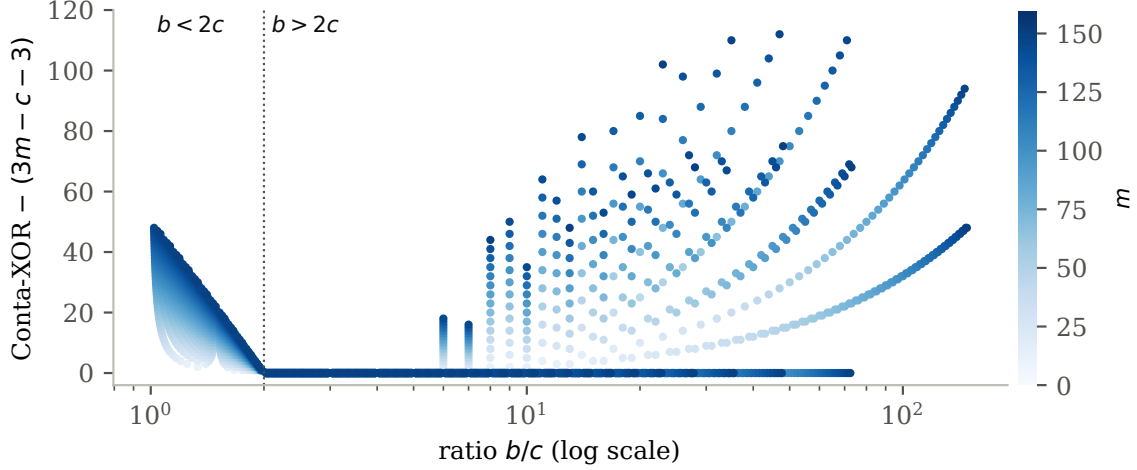


Figure 2: ♦ Greedy count minus closed form, $CX(f) - (3m - c - 3)$, for the 3 541 generic members with $m \leq 150$. The difference is never negative. Left of $b/c = 2$ it equals $2c - b$; between 2 and 6 it is zero; for larger ratios the greedy algorithm loses gates on members with small c .

Table 2: ♦ Irreducible members of the second family at the NIST degrees. “Thesis” marks the values published in [3, Tab. 17].

m	f	b	c	dXOR	CX	$3m - c - 3$	depth
163	$x^{163} + x^{117} + x^{71} + x^{46} + 1$	71	46	649	461	440	4
233	$x^{233} + x^{158} + x^{83} + x^{75} + 1$	83	75	866	688	621	4
233	$x^{233} + x^{217} + x^{201} + x^{16} + 1$	201	16	1687	680	680	7
409	$x^{409} + x^{294} + x^{179} + x^{115} + 1$	179	115	1642	1160	1109	4
409	$x^{409} + x^{337} + x^{265} + x^{72} + 1$	265	72	1823	1152	1152	4
571	$x^{571} + x^{389} + x^{207} + x^{182} + 1$	207	182	2145	1685 (thesis)	1528	4
571	$x^{571} + x^{465} + x^{359} + x^{106} + 1$	359	106	2531	1604 (thesis)	1604	4
571	$x^{571} + x^{549} + x^{527} + x^{22} + 1$	527	22	6297	1688 (thesis)	1688	11

Randomising ties improved the greedy count in 212 generic members, all with $b \geq 6c$ and by at most 12 XORs, but closed the gap to the closed form in only one of them. For special members the greedy algorithm can be shorter (for instance $\frac{12}{5}m - 2$ when $b = 3c$), as expected for polynomials of the form $G(x^9)$ with a short G .

Figure 2 shows the difference $CX(f) - (3m - c - 3)$ as a function of b/c . The observation explains the numbers in the thesis. Table 2 lists the members of the second family that appear in [3, Tab. 17] and those at the other NIST degrees: when $b > 2c$ the thesis value coincides with $3m - c - 3$, and when $b < 2c$ the closed form removes $2c - b$ gates. The irreducible member $x^{163} + x^{117} + x^{71} + x^{46} + 1$ has $b < 2c$; its greedy count is 461 and the closed form gives 440.

Small degrees. For the generic members of degrees 9 and 10, $f_{5,2}$ and $f_{4,3}$, the Boyar–Peralta heuristic [8], which may use cancellations (best of 10 runs), finds programs of lengths 22 and 24, equal to $3m - c - 3$; the greedy counts are 22 and 26. For the smallest member $f_{3,2}$ (degree 7, special) both the closed form and Boyar–Peralta give 16, one less than the greedy count.

8 Comparison with the first family and with other polynomials

Choosing c . By Theorem 6 the cost decreases with c . Since $c \leq \lfloor (m - 1)/3 \rfloor$, the best possible generic member has cost close to $3m - \frac{m}{3} = \frac{8}{3}m$. For the first family the cost $3m - 2$ does not

Table 3: ♦ Census for $5 \leq m \leq 1100$ (1 096 degrees). “Best” uses [Theorem 6](#) for the second family, $3m - 2$ or $\frac{12}{5}m - 1$ for the first family [2], and $\frac{7}{4}m - 1$ for equally spaced members.

irreducible members of the second family	1 508
degrees with at least one member: second family / first family	720 / 502
degrees with members of both families	405
of which the second family is cheaper	403 (mean saving 7.3%)
of which the first family is cheaper / tie	1 ($m = 155$) / 1 ($m = 5$)
degrees without irreducible trinomials	502
covered by the second family / by the first family	288 / 229
covered by the second family only	112
covered by neither	161
mean cost per bit of the best generic member of the second family	2.77

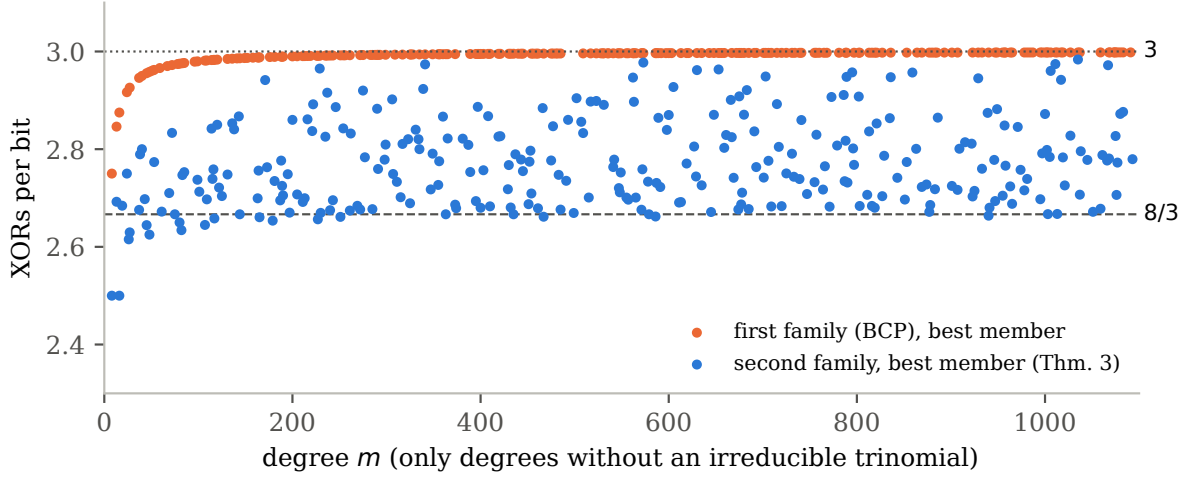


Figure 3: ♦ Best reduction cost per bit, for the degrees $m \leq 1100$ without irreducible trinomials, of the best member of the first family (BCP) and of the second family ([Theorem 6](#)). The dashed line is the asymptotic limit $\frac{8}{3}$ of the second family.

depend on (b, c) , except in the special case $m = 5c$.

Census. We tested the irreducibility of all members of both families and of all trinomials for $5 \leq m \leq 1100$, using Rabin’s test [24]. [Table 3](#) summarises the result, and [Fig. 3](#) shows, for the degrees without irreducible trinomials, the cost per bit of the best member of each family.

The only degree in which the first family wins is $m = 155$, where its member with $b = 2c$ reaches $\frac{12}{5}m - 1 = 371$ and the best generic member of the second family needs 411. Degrees $\equiv 0 \pmod{8}$ deserve a comment, since they never have irreducible trinomials; the second family has members in 25 of them, for instance $x^{16} + x^{11} + x^6 + x^5 + 1$ ([Example 9](#)), $x^{48} + x^{33} + x^{18} + x^{15} + 1$ (126 XORs) and $x^{144} + x^{99} + x^{54} + x^{45} + 1$ (384 XORs).

NIST degrees. [Table 4](#) compares the best members of both families at the NIST degrees with the NIST polynomials and with trinomials where they exist. At $m = 163$ and $m = 571$, where no irreducible trinomial exists, the second family reduces with 22.9% and 23.7% fewer XORs than the NIST pentanomials (as counted by Conta-XOR) and 9.7% and 10.7% fewer than the first family. At $m = 283$ the second family has no irreducible member. At 233 and 409 trinomials are cheaper, as expected.

Table 4: ♦ Best known reduction costs at the NIST degrees. NIST pentanomials: Conta-XOR counts from [3]; first family: [2]; second family: Theorem 6.

m	NIST polynomial	trinomial	first family	second family
163	571	–	487	440
233	464 (trinomial)	464	697	621
283	862	–	847	–
409	816 (trinomial)	816	–	1109
571	2003	–	1711	1528

9 Discussion and conclusion

Why the second family was hard. For the first family, $x^m \equiv (x^b + 1)(x^c + 1)$ and $m = 2b + c$ make the high part fold into the low part after a bounded number of steps. For the second family the quotient has long-range dependencies of stride $3c$, visible only after solving the recurrence (4). The greedy counter captures part of this structure, and it reaches the optimum of our construction when $2c < b < 6c$, but it misses the sharing across the comb sums when $b < 2c$ and loses track of the long combs when $b \gg c$. This explains the remark in the thesis that the family required “a greater understanding of the reduction process”.

What the closed form gives. The program of Algorithm 1 is uniform, easy to generate for any (b, c) , has small depth when $b \leq 4c + 1$, and its cost depends on the parameters only through m and c . It turns the second family into a practical alternative wherever it has irreducible members, and in particular in degrees without trinomials.

Limitations and open questions. (i) Theorem 6 is an upper bound. For the generic members with $m \leq 10$ it matches the Boyar–Peralta heuristic, but lower bounds for larger degrees are open. (ii) Special members, of the form $G(x^g)$ with a short G , can be reduced with fewer gates; a uniform treatment of these cases is left open. (iii) The irreducibility of the members is still characterised only computationally. (iv) Joint optimisation of the product and the reduction, as in Mastrovito-type multipliers [20, 25], and area–delay trade-offs remain to be studied.

Reproducibility. The generator of Algorithm 1 with symbolic verification, the irreducibility census and the scripts that produced the tables and figures are available from the authors and will be published with the final version.

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